



面向服饰纹理替换的智能重绘 方法研究与系统开发

张雅瑞
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01

选题意义与可行性分析

背景：

- 围绕在电商和虚拟试衣领域， 用户和设计师的需求

痛点：

- 传统方法效率低
- 现有AI缺陷

○ 课题意义与可行性

课题目标：

开发一套“智能重绘框架 (iRetexturing)”的算法和框架。

应用价值：

- 为设计师提供一站式原型工具；
- 优化电商用户的购物决策

可行性：

- SDXL、ControlNet和SAM等成熟算法框架
- React和Next.js的系统前后端架构



02

研究的基本内容

○ 四大研究内容

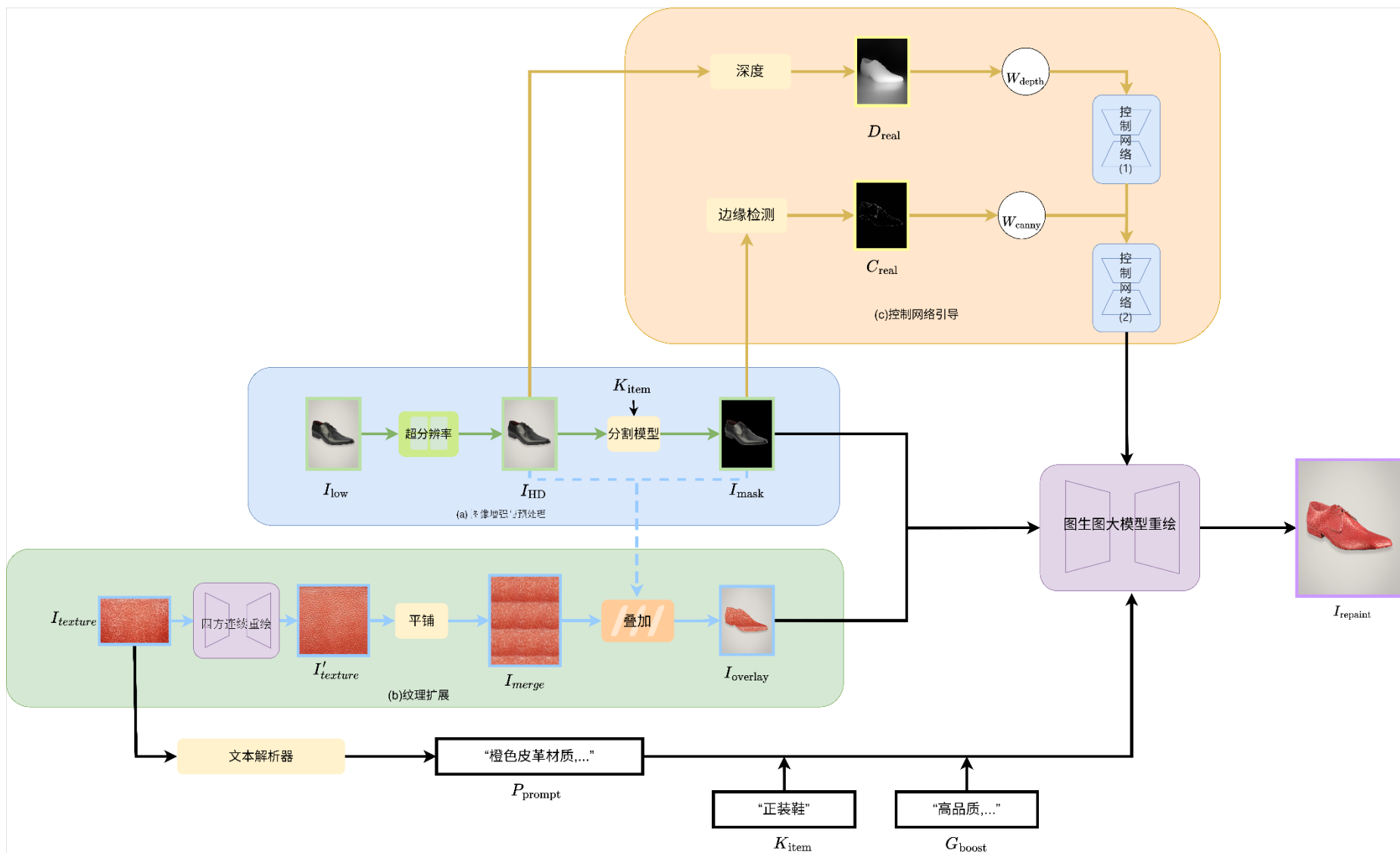
- **几何保真算法：** 保持衣服的褶皱和立体结构
- **无缝纹理扩展：** 解决大面积花色平铺时的接缝问题
- **交互式前端：** 可视化的生图控制界面
- **高负载后端：** 高分辨率生成任务和数据库管理系统



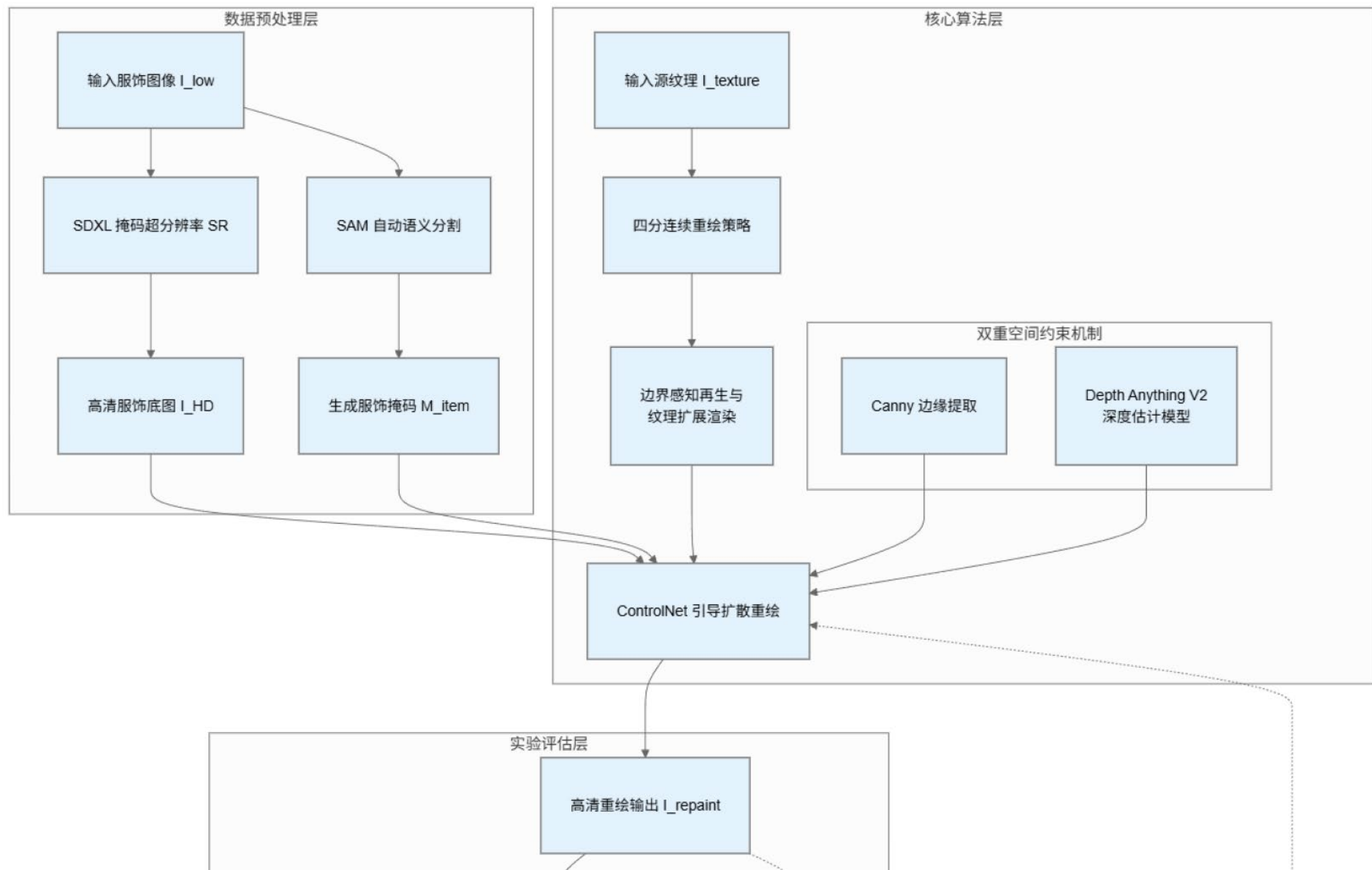
03

总体技术路线

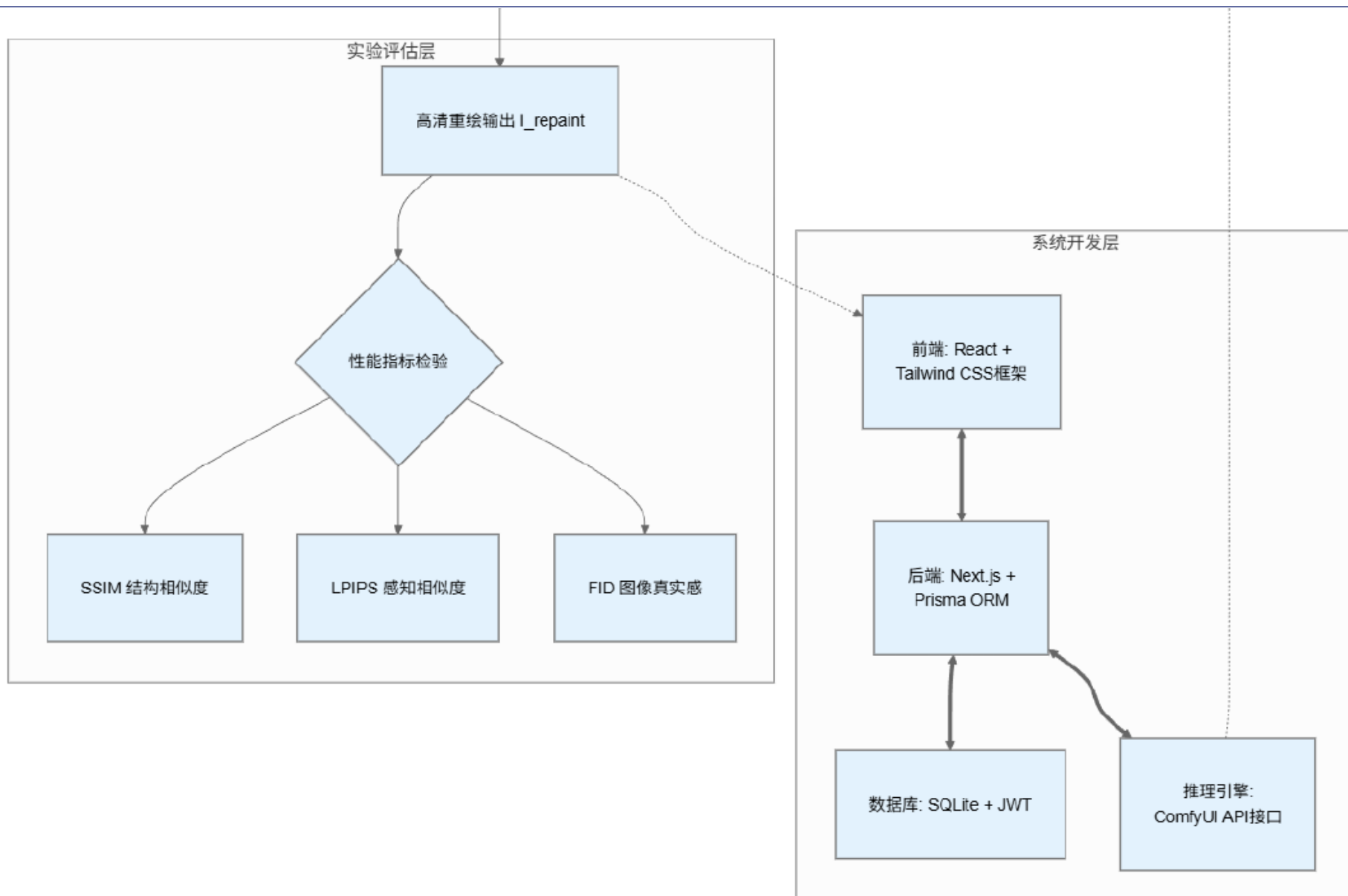
技术路线



关键算法讲解



关键算法讲解

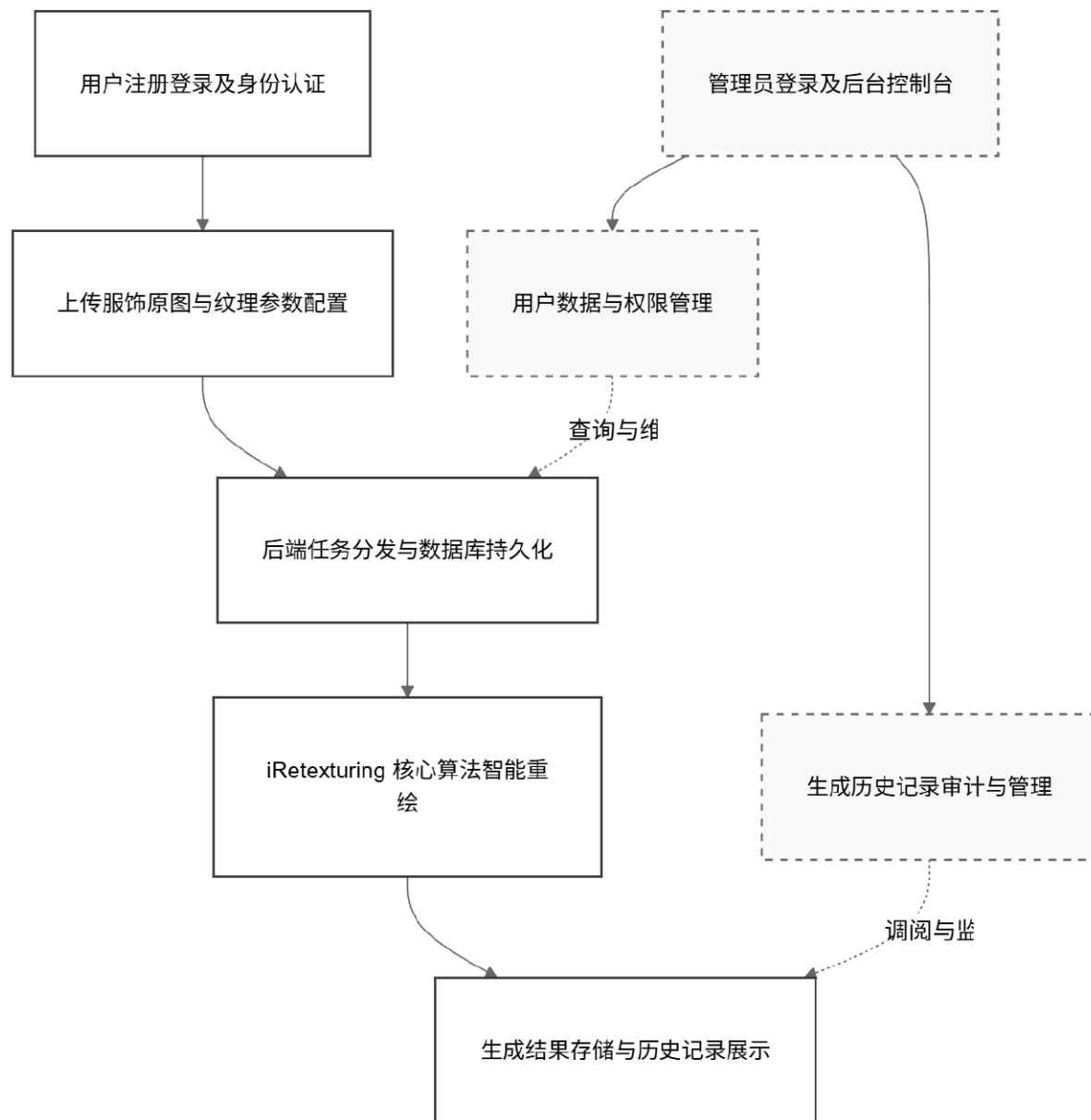




04

系统架构与实现

系统架构设计



前端： 使用 React + Tailwind CSS

后端： 使用 Next.js + Prisma ORM + SQLite

推理引擎： 通过API调用本地部署的 ComfyUI 工作流



05

功能演示

功能演示

The screenshot displays the ViewComfy IDE interface. The left sidebar shows a file explorer with a project structure including files like `services`, `stores`, `types`, `.dockerignore`, `.env`, `.eslintrc.json`, `.gitignore`, `测试api.json`, `换纹理-V13-轻量化api.json`, `换纹理-V13-全量api.json`, `AGENTS.md`, `components.json`, `debug-backend.js`, `Dockerfile`, `eslint.config.mjs`, `instrumentation-client.ts`, `instrumentation.ts`, `LICENSE`, `middleware.ts`, `next-env.d.ts`, `next.config.mjs` (selected), `package-lock.json`, `package.json`, `postcss.config.mjs`, `prisma.config.ts`, `README.md`, `run_comfy_heavy_only.bat`, `run_heavy.bat`, `sentry.edge.config.ts`, `sentry.server.config.ts`, `server.log`, `styles.css`, `test-auth.js`, `tsconfig.json`, `tsconfig.tsbuildinfo`, `view_comfy-轻量化.json`, `view_comfy-全量api.json`, `view_comfy-运行测试.json`, `view_comfy.json`, and `ViewComfy-modal.dockerfile`. The main editor shows the `next.config.mjs` file with the following content:

```
1 import {withSentryConfig} from "@sentry/nextjs";
2 /** @type {import('next').NextConfig} */
3 const nextConfig = {
4   images: { remotePatterns: [{ hostname: "*" }] },
5   output: "standalone",
6   devIndicators: {
7     // 隐藏开发环境的路由/状态提示
8     autoPrerender: false
9   }
10 };
11
12 export default withSentryConfig(nextConfig, {
13   // For all available options, see:
14   // https://www.npmjs.com/package/@sentry/webpack-plugin#options
15
16   org: "viewcomfy",
17
18   project: "viewcomfy-front-end",
19
20   // Only print logs for uploading source maps in CI
21   silent: !process.env.CI,
22
23   // For all available options, see:
24   // https://docs.sentry.io/platforms/javascript/guides/nextjs/manual-setup/
25
26   // Upload a larger set of source maps for prettier stack traces (increases build time)
27   widenClientFileUpload: true,
28
29   // Route browser requests to Sentry through a Next.js rewrite to circumvent ad-blockers.
30   // This can increase your server load as well as your hosting bill.
31   // Note: Check that the configured route will not match with your Next.js middleware, otherwise reporting
32   // side errors will fail.
33   tunnelRoute: "/monitoring",
```

The right sidebar shows the `JS javascript` file with the following content:

```
// For all available options, see:
// https://www.npmjs.com/package/@sentry/webpack

org: "viewcomfy",

project: "viewcomfy-front-end",

// Only print logs for uploading source maps in
silent: !process.env.CI,

// For all available options, see:
// https://docs.sentry.io/platforms/javascript/

// Upload a larger set of source maps for prett
widenClientFileUpload: true,

// Route browser requests to Sentry through a N
// This can increase your server load as well a
// Note: Check that the configured route will n
// side errors will fail.
tunnelRoute: "/monitoring",

// Automatically tree-shake Sentry logger state
disableLogger: true,

// Enables automatic instrumentation of Vercel
// See the following for more information:
// https://docs.sentry.io/product/crons/
// https://vercel.com/docs/cron-jobs
automaticVercelMonitors: true,
});
```

The bottom terminal shows the output of the `npm i baseline-browser-mapping@latest -D` command, indicating that the `"middleware"` file convention is deprecated and should be replaced with `"proxy"`. The terminal also shows the `Starting...` message and the `Ready in 1333ms` status.



感谢您的观看